Grade 8
Unit 5
Lesson 7 Practice

Name____________________
Date____________________
Period____________________

William is a window washer and needs to lean his ladder against the house to reach the windows at the top. He sets the ladder 8 feet from the house that is 14 feet tall. What is the minimum length ladder he needs to use to reach the top of the house?
Round your answer to the nearest tenth.

1. Model the problem.

2. State the Pythagorean theorem and substitute the values.

3. Solve for the unknown.

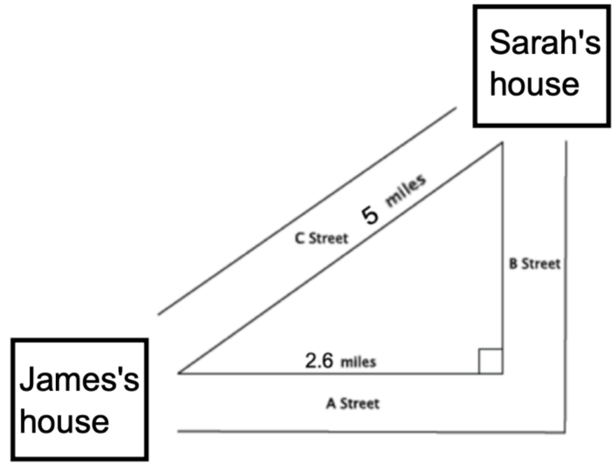
When you purchase a T.V., the size refers to the diagonal measurement of the screen. If the 46 inch TV has a square screen, what is the approximate length and width? Round your answer to the nearest tenth.

4. Model the problem.

5. State the Pythagorean theorem and substitute the values.

6. Solve for the unknown.

There are two paths that one can use to go from Sarah's house to James' house. One way is to take C Street, and the other way requires you to use A Street and B Street. How much shorter is the direct path along C Street?



7. What is the distance of B street?
Round your answer to the nearest tenth.

8. How much shorter is the direct path along C Street?

An Airplane left Orlando International airport and flew south for 720 miles and then went east for a certain distance. The plane ended up $2{,}442$ miles from where it started. How far did the plane fly east?

9. David set up the equation $2442^2 + b^2 = 720^2$ to solve for the missing distance. Do you agree or disagree with David's equation?

10. Correct David's equation and solve to determine how far the plane flew east? Round your answer to the nearest tenth.